

MLEvolve AWS GPU Allocation Request

For Professor Xie - a continuous allocation supported by reproducible scheduling

Request: 1 A100 80 GB + 1 A10

Summary

A paired replay of the same V100 Cassava trace shows a modest, reproducible scheduling gain after removing the fixed concurrency cap: makespan decreases by 3.03% and p95 flow time by 21.52%. The replay also exposes a fairness tradeoff in worst-case waiting. The requested allocation will validate this result end-to-end on matched hardware.

Paired profile-based replay: same V100 Cassava trace

Method	Original MLEvolve	Current scheduler
Makespan	314.17 min	304.65 min (-3.03%)
p95 flow time	62.12 min	48.75 min (-21.52%)
Mean flow time	22.79 min	22.40 min (-1.70%)

Tradeoff: max wait increases from 33.68 to 53.92 min and starvation count from 1 to 4. Both policies use the same 39 schedulable nodes, releases, profile memory values, measured slowdown, and 31 GB budget; parallel cap is unset.

No fixed Max Parallel Jobs: branch profiles, live VRAM, and incremental admission select concurrency.

Real PetFinder evidence: capacity need and feasibility

Original MLEvolve encountered a real CUDA Out-of-Memory error during concurrent admission. This is the unsafe admission event profile-based scheduling is intended to avoid.

The controlled Scheduler comparison is unfinished and is not used as a performance claim.

Requested continuous AWS allocation

1 x NVIDIA A100 80 GB (or H100 80 GB)

Shared Qwen3.8-27B INT8 vLLM Agent: local serving was validated before the controlled experiment.

1 x NVIDIA A10 24 GB

One controlled worker: same-hardware 50-node baseline/Scheduler comparison, ablations, and sequential repeats.

Separating Agent and training prevents latency, VRAM contention, and OOM from jointly contaminating scheduler metrics.

Evidence: scheduler_comparison.json; paired profile replay code; PetFinder A100-A10 record and baseline Gantt.

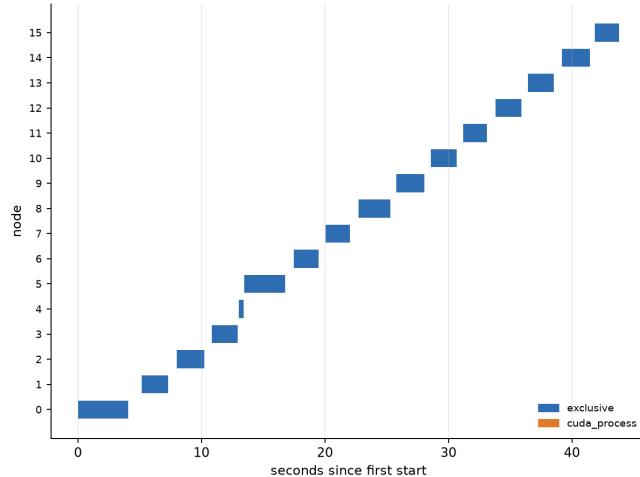
Rigor: replay is deterministic, paired, and profile-based; it is not a substitute for a live matched-hardware run.

Prior empirical evidence: runtime profiles work on real GPUs

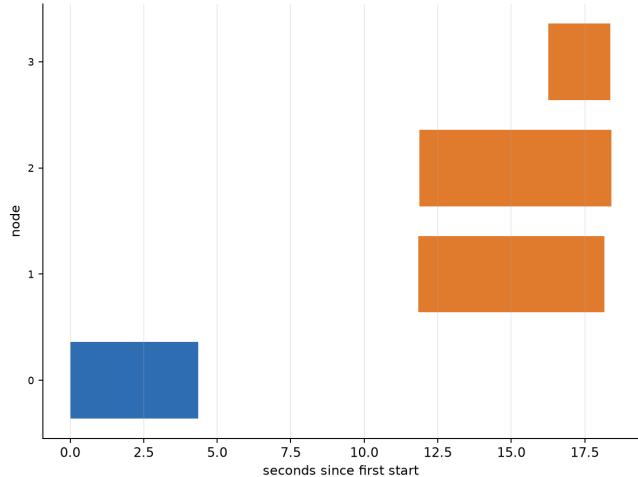
Supporting evidence for this request; completed results are not replaced by unfinished controlled comparisons

RTX 5090 scheduler validation: Gantt and metric nodes

Cold-start: distinct signatures



Profiled: shared signature



Cold-start: distinct signatures

completed	16/16
placement	exclusive=16
mean train	0.836s
mean runner overhead	0.552s
mean dispatch latency	17.340s
GPU avg / peak SM	15.1% / 99%
GPU peak VRAM	11152 MiB
OOM	0

Profiled: shared signature

completed	4/4
placement	cuda_process=3, exclusive=1
mean train	3.666s
mean runner overhead	0.729s
mean dispatch latency	1.158s
GPU avg / peak SM	32.2% / 99%
GPU peak VRAM	13401 MiB
OOM	0

Figure 1. RTX 5090: cold-start stays conservative; a shared signature admits 3 CUDA-process jobs after profiling.

Real run: profile-guided admission

The profiled run executes an anchor, then incrementally admits CUDA-process work. No fixed parallel-job cap is configured.

PetFinder: estimate, training, calibration

An RTX 5090 PetFinder run wrote its branch profile during execution and calibrated it with observed wall-clock time for matching future jobs.

Existing Predictor and Agent assets

Prepared

A10 profile-label corpus

Covered

batch-size calibration

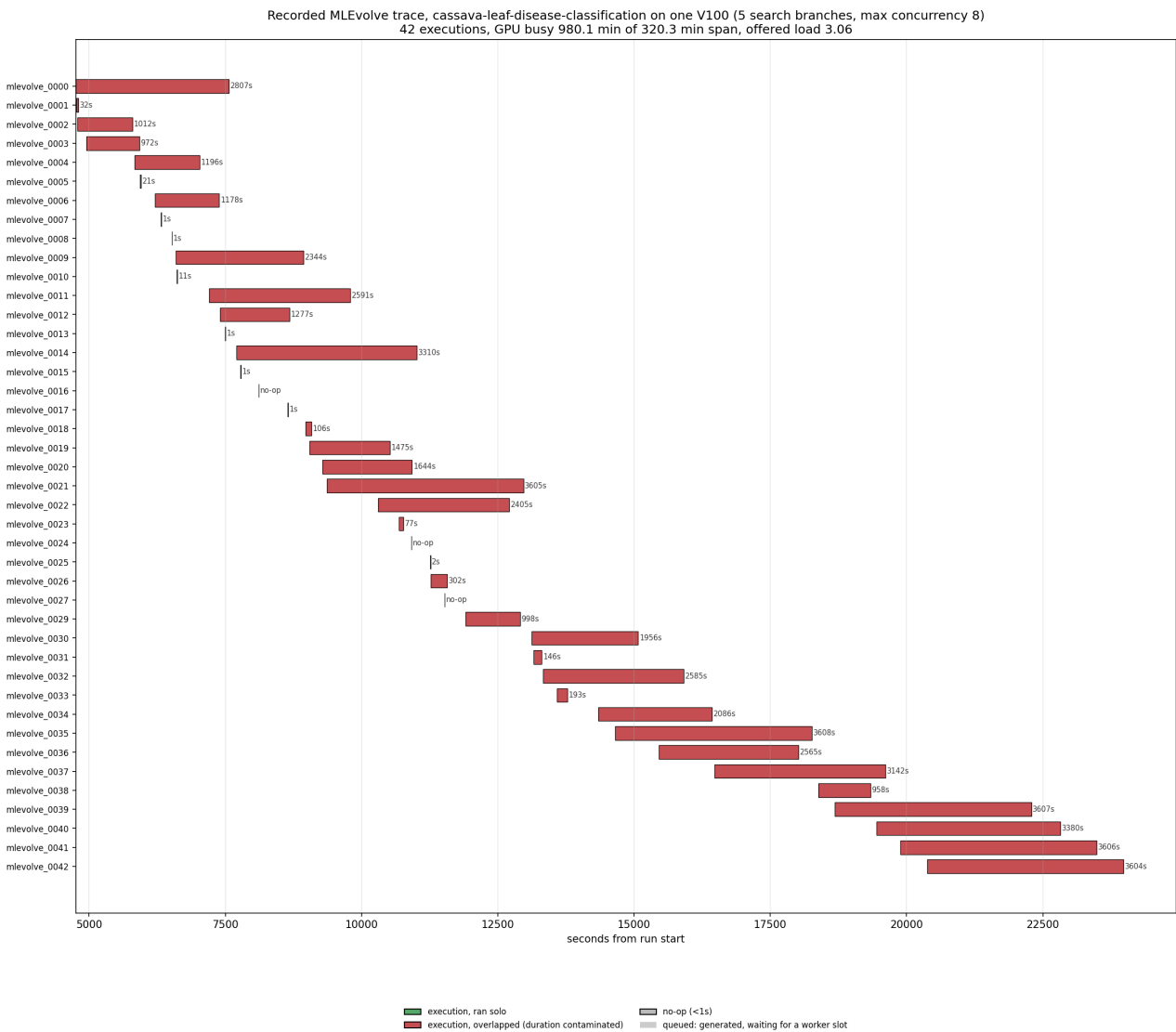
Validated

local A100 vLLM Agent

A10 labels cover model, precision, batch size, VRAM, and epoch time. The local A100 vLLM Agent was validated separately. Together these assets support separating the Agent from the single controlled training worker.

Cassava on one V100: DeepSeek trace

Original Gantt artifact retained for audit; it is not used for a causal speedup claim.



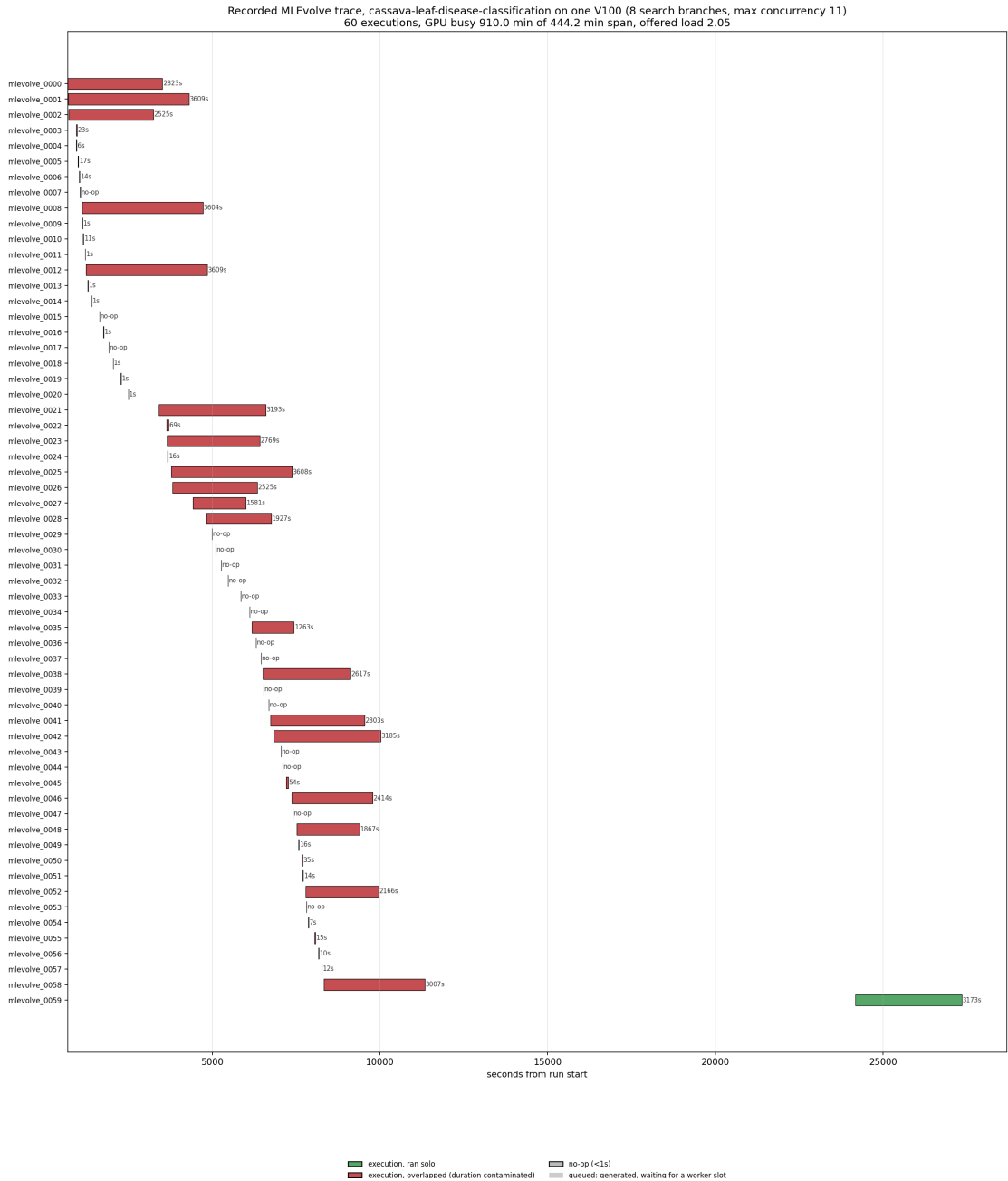
Trace record: 42 executions and 18 successful nodes. The figure is retained as an auditable run artifact.

No speedup or causal comparison is claimed from this agent-specific trace.

Any future comparison will use a shared completed-node count and the same hardware and workload configuration.

Cassava on one V100: Sonnet 5 trace

Original Gantt artifact retained for audit; it is not used for a causal speedup claim.



Trace record: 60 executions and 15 successful nodes. The figure is retained as an auditable run artifact.

No raw-span, speedup, or causal comparison is claimed from this agent-specific trace.